Storage and prediction excerpt (continued) *storage/flight_data_store.py*

```
def _haversine_m(lat1, lon1, lat2, lon2):
    r = 6371000.0
    p1 = math.radians(lat1)
    p2 = math.radians(lat2)
    dlat = math.radians(lat2 - lat1)
    dlon = math.radians(lon2 - lon1)
    a = math.sin(dlat / 2) ** 2 + math.cos(p1) * math.cos(p2) * math.sin(dlon / 2) ** 2
    return 2 * r * math.atan2(math.sqrt(a), math.sqrt(1 - a))


def _bearing_deg(lat1, lon1, lat2, lon2):
    p1 = math.radians(lat1)
    p2 = math.radians(lat2)
    dlon = math.radians(lon2 - lon1)
    y = math.sin(dlon) * math.cos(p2)
    x = math.cos(p1) * math.sin(p2) - math.sin(p1) * math.cos(p2) * math.cos(dlon)
    deg = (math.degrees(math.atan2(y, x)) + 360.0) % 360.0
    return deg


class FlightTrajectoryStore:
    """
    Compact training-data store.
    - flight_latest: matrix-like table (one row per flight, fields in columns).
    - trajectory_samples: time-series points for trajectory prediction.
    """

    def __init__(
        self,
        db_path,
        enabled=True,
        min_interval_sec=5,
        min_distance_m=30,
        min_altitude_change_m=20,
        min_heading_change_de2=500,
        min_speed_change_cms=200,
        history_retention_days=30,
        append_csv_enabled=True,
        append_csv_dir=None,
        icao_excel_enabled=True,
        icao_excel_dir=None,
        icao_master_enabled=True,
        icao_master_path=None,
        reset_excel_on_start=True,
        icao_excel_extension="xlsx",
    ):
        self.enabled = bool(enabled)
        self.db_path = Path(db_path)
        self.min_interval_sec = int(min_interval_sec)
        self.min_distance_m = float(min_distance_m)
        self.min_altitude_change_m = int(min_altitude_change_m)
        self.min_heading_change_de2 = int(min_heading_change_de2)
        self.min_speed_change_cms = int(min_speed_change_cms)
        self.history_retention_days = int(history_retention_days)
        self.append_csv_enabled = bool(append_csv_enabled)
        self.append_csv_dir = Path(append_csv_dir) if append_csv_dir else self.db_path.parent / "training_csv"
        self.icao_excel_enabled = bool(icao_excel_enabled) and _OPENPYXL_AVAILABLE
        self.icao_excel_dir = Path(icao_excel_dir) if icao_excel_dir else self.db_path.parent / "icao_excel"
        ext = str(icao_excel_extension or "xlsx").strip().lower().lstrip(".")
        self.icao_excel_extension = ext if ext in ("xls", "xlsx") else "xlsx"
        self.icao_master_enabled = bool(icao_master_enabled) and self.icao_excel_enabled
        self.reset_excel_on_start = bool(reset_excel_on_start)
        self.icao_master_path = Path(icao_master_path) if icao_master_path else None
        self._master_wb = None
        self._master_headers = [
            "timestamp",
            "ICAO",
            "callsign",
            "latitude",
            "longitude",
            "altitude",
            "heading",
            "speed",
            "vertical_rate",
```